

P08 - AI Research Assistant

Group Formal Tag:	SOFT3888_TU08_01
Tutor Name:	Michael de Francesco
Client Name:	Vecton AI (Bocheng Han)

Group members details

No.	Student Name	Student ID	Student Unikey	XP-Roles
1	Joshua Morton	530500735	jmor4801	tracker
2	Prithish Lahiri	530609175	plah4259	programmer
3	Fiona Lee	540880966	flee0409	Manager
4	Kang Le Charles Lim	540375026	klim0039	Tester
5	Felicia Chen	530532079	lche6131	Programmer
6	Xiyuan Pan	530667566	xpan0988	Doomsayer
7	Xingyu Liu	510015523	xliu3209	Customer liason

Attendances: *All Attended*

Apologies: *N/A*

Submission Date:6th

Submission Date: 1/09/26

Weekly Group Meeting Minutes

Time: *1st Meeting of week 5 in tutorial), 1st September*

Venue: *Tutorial room (J02.03.315 PNR Building Learning Studio 315)*

Meeting Minute Taker: *Prithish Lahiri*

Attendances: *Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu*

Apologies: *none*

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	The frontend skeleton, GraphRAG module, and Supabase/PostgreSQL database are all operational.	Consolidate individual components ahead of integration.	Everyone	No immediate escalation
2	What is in progress?	Data is being gathered in order to reach the 150-profile goal; work on GraphRAG and search/ranking algorithms is still ongoing.	Continue gathering data and complete the GraphRAG/search logic.	Everyone	No immediate escalation
3	What is working well?	The frontend, database, and GraphRAG components were all developed separately and are operating as planned.	Continue at the current speed throughout the integration phase.	Everyone	No immediate escalation
4	What needs improvement?	It is necessary to integrate separate components into a single functional system.	After data collection is finished, schedule an integration sprint.	Everyone	No immediate escalation
5	Reminders	Every member attested to the tasks that must be completed by the end of the week.	Set completion goals for each person.	Everyone	No immediate escalation
6	What needs to be completed before the next meeting?	Make major progress in data collection and GraphRAG/search algorithm finalisation.	Give these two things top priority.	Everyone	No immediate escalation

7	What's Next	Integrate all sections after combining the data collection output, then start testing, debugging, and refining.	Enter the phase of integration and testing.	Everyone	No immediate escalation
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Main Discussions

- The team verified that the database, frontend, and GraphRAG are all functionally complete on their own.
- The remaining tasks were discussed, including completing the search/ranking algorithm and data collection before integration can start.

Key Takeaways

- The project is now in the integration phase, having completed the individual-component development phase.
- The two remaining obstacles before complete system testing are data collection and the GraphRAG/search algorithm.

What Happened

- Database, frontend, and GraphRAG development were carried over from last week and are now finished.
- Work on data collection and search/ranking algorithms is still ongoing and will be incorporated into this week's assignments.

What's Next

- Joshua and Fiona is responsible for finalising data collection toward the 150-profile target.
- Xiyuan is responsible for finalising the GraphRAG implementation.
- Felicia is responsible for finalising the search/ranking algorithm.
- Prithish is responsible for finalising the frontend
- Charles is responsible for working on the database
- Everyone is responsible for preparing for the upcoming integration and testing phase.

Weekly Client Meeting Minutes

Time: 3rd September 4:30pm

Venue: Google Meet

Meeting Minute Taker: Prithish Lahiri

Attendances: *Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu*

Apologies: *none*

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	The team updated all nine universities on the proposal, GraphRAG module, frontend skeleton, and data collection strategy.	The client reviewed the progress and found it satisfactory.	Everyone	No immediate escalation.
2	What is in progress?	Search/ranking algorithm, GraphRAG refinement, and data collection.	Proceed as scheduled.	Everyone	No immediate escalation.
3	What is working well?	The frontend structure and GraphRAG/search components are already developed, which is good, according to the client.	Maintain the current strategy.	Everyone	No immediate escalation.
4	What needs improvement ?	The client noted that author-name disambiguation requires special attention because inaccurate matching compromises the reliability of the profile and summary. It should be mentioned that the database needs to manage duplicates and conflicts and preserve source information and collection dates.	Give data provenance tracking and disambiguation accuracy top priority.	Fiona, Joshua	Author disambiguation quality to be monitored closely.

5	Reminders	The client affirmed that the risk associated with GraphRAG complexity is reasonable and supports the flat-RAG fallback in the event that it turns out to be more dependable. As best-effort, non-CS/IT coverage was confirmed to be acceptable.	Give functional completeness, accuracy, and citation quality precedence over technical complexity.	Everyone	No immediate escalation.
6	What needs to be completed before the next meeting?	Continue gathering data and finalising GraphRAG and searches while taking client input into account.	Take into account customer comments regarding data provenance and disambiguation.	Everyone	No immediate escalation.
7	What's Next	After the remaining components are finished, proceed to integration and testing.	Start preparing for integration.	Everyone	No immediate escalation.

Main Discussions:

- The client was guided through each section's progress by the team, including the proposal, database, frontend, GraphRAG, and data collection.
- The client confirmed that they were satisfied with the current course and identified two particular priorities: data provenance (source/date/duplicate handling) and author disambiguation accuracy.

Key Takeaways:

- The client is happy with the technical approach and current progress, including the flat-RAG backup plan.
- The client clearly prioritises functional completeness, accuracy, and citation quality over technical complexity; this is helpful advice if trade-offs occur during integration.
- Data provenance tracking and author-name disambiguation were identified as areas that require special attention.

What Happened:

- The client reviewed and verified that the project's direction is still in line with the initial requirements after the team presented progress across all workstreams.
- The client gave detailed input on data quality and disambiguation that should be used moving forward.

Whats Next:

- Fiona is in charge of handling duplicate or conflicting records and making sure the database maintains source information and collection dates.
- Joshua is in charge of giving author-name disambiguation extra consideration when gathering data.
- It is everyone's responsibility to keep moving with the work they were assigned and work in the direction of integration while incorporating this feedback.

Out of Tutorial meeting

Time: *3rd Meeting of week 4 (Out of tutorial group meeting), 6th September, 4:00pm*

Venue: *Instagram Call (25mins)*

Meeting Minute Taker: *Prithish Lahiri*

Attendances: *Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu*

Apologies: *none*

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	The frontend, GraphRAG module, and Supabase/ PostgreSQL database were all verified to be finished and operational.	Proceed with integration.	everyone	No immediate escalation.
2	What is in progress?	Work on search/ranking algorithms and data collection is still in progress.	Complete prior to the start of integration.	everyone	No immediate escalation.
3	What is working well?	Every individually owned component was finished on its own without any significant problems.	Keep the momentum going.	everyone	No immediate escalation.
4	What needs improvement?	Before integration can begin, the remaining components (data collection, search algorithm) must be completed.	Establish definite completion goals for each.	everyone	No immediate escalation.

5	Reminders	Before integration, the team kept each other informed about the status of each individual section.	Verify preparedness before assembling parts.	everyone	No immediate escalation.
6	What needs to be completed before the next meeting?	Complete the search/ranking algorithm and data collection; start merging all sections.	Make integration readiness a top priority.	everyone	No immediate escalation.
7	What's Next	After integrating all sections and combining the data collection output, start testing, debugging, and polishing.	Enter the phase of integration and quality assurance.	everyone	No immediate escalation.

Main Discussions

- The team verified that GraphRAG, the frontend, and the database are all finished and prepared for integration.
- The remaining gap was discussed: before full-system integration can start, data collection and the search/ranking algorithm must be completed.

Key Takeaways

- The project is on schedule to transition from the development of independent components to integration.
- The last obstacles before testing and debugging can begin are data collection and search/ranking algorithms.

What Happened

- Database, frontend, and GraphRAG development are at a comfortable place, as a result of this week's earlier meetings.
- The search/ranking algorithm and data collection are still ongoing and continue into the integration stage.

Whats Next:

- Completing the data collection at each of the nine universities.
- Finalising the search/ranking algorithm
- Once these two components are finished, everyone is in charge of starting the integration of all sections, which is followed by testing, debugging, and polishing.

Individual reflections:

Name: Joshua Morton

Student ID: 530500735

Goals

- + Begin data collection for our project
 - + Ensure that the schema is polished and that the majority of data required is identified within it
 - + Create documentation for the database schema
 - + Access the OpenAlex API data, filter out the data which is not needed, and compile all required data into a consistent format
- + Begin integrating the database into the project so that it can be used for all other functions

Progress Made

All goals were achieved. A script was developed to systematically extract data from Computer Science research papers from Australian universities. It is currently limited to 1200 papers (12 universities, 100 papers each), however, can very easily be ramped up to 100,000 papers/day.

The database was also started to be integrated into project.

Blockers to Progress/Issues

Busyness with other university assignments and work.

Name: Prithish Lahiri

Student ID: 530609175

Goals:

- ⇒ Create a working front-end framework (auth, home, directory search, profiles, Ask AI) that colleagues can enter actual data into.
- ⇒ To get ready for backend/database integration, set up a clear, interchangeable API layer.

Progress Made:

- ⇒ Created and released a fully functional frontend skeleton that included routing, mock-data-driven functionality, and all seven core pages.
- ⇒ AI Q&A with citations, filterable directory search, and authentication flow were implemented. A demo-ready user interface was achieved by applying
- ⇒ Full visual theming and polish (dark theme, hover interactions, loading states).

Blockers to Progress/Issues:

- ⇒ Currently using placeholder schema assumptions; final Supabase table/column names have not yet been confirmed with a database teammate.

⇒ The frontend is currently using mock data because the real backend API (GraphRAG/LLM integration) is not yet accessible.

Name: Fiona Lee

Student ID: 540880966

Goals

- Collect data (researchers, papers, etc.) from university websites
- Collect data in spreadsheet with corresponding columns from database schema

Progress Made

- Collected 100 researcher profiles from university websites
- Collected research papers for 1 university
- Integrated database with fellow teammates

Blockers to Progress/Issues

- Decision of which LLM to use is postponed
- Data needs to be collected from different sites and requires extra work to do

Name: Felicia Chen

Student ID: 530532079

Goals

Develop the ranking section based on last week's proposal.

Progress Made

Communicated with Xiyuan, who is responsible for RAG development. And we confirmed the general process and framework from retrieval candidate papers to ranking them based on different factors and then generating results. Before the database was set up, I developed and tested with mock data to evaluate the scores and rankings of candidate papers under different search preferences. The output is acceptable. The next step is to integrate the RAG and ranking section with Xiyuan. Xiyuan will integrate the database with RAG, while I will integrate the ranking section with the front end(Lahiri).

Blockers to Progress/Issues

Xiyuan and I have some differing opinions on the ranking factors. For example, he suggested a graph relevance factor that would require establishing an academic network to find connections between academics and further create a paper network to record the relationships between papers. I believe this approach has a narrow application scope, as it mainly benefits users searching for situations like 'I want to

see the publications of Felicia's collaborators.' The resources consumed would be too high, which could reduce performance. I recommend focusing more on RAG instead, so I decided to abandon this factor.

Name: Kang Le Charles Lim

Student ID: 540375026

Goals

My goals for this week are to set up Supabase and sql codes for the database.

Progress Made

- Set up Supabase
- Finish coding the sql for the database

Blockers to Progress/Issues

There were no issues this week.

Name: Xiyuan Pan

Student ID: 530667566

Goals

- Complete the initial GraphRAG backend.
- Prepare integration with the database and ranking module.
- Coordinate ranking factors and interface design with Felicia.

Progress Made

- Implemented the standalone GraphRAG backend with retrieval, graph traversal, evidence handling, and fallback logic.
- Added mock data, in-memory components, documentation, and 14 unit tests.
- Discussed the GraphRAG-to-ranking workflow and ranking factors with the team.
- Planned the next integration stage with the database and frontend.

Blockers to Progress/Issues

- Final database schema is still being consolidated.
- Ranking formula and GraphRAG-ranking interface still need testing with real data.
- Free-tier LLM/embedding providers have not yet been finalized.

Name: Xingyu Liu

Student ID:

Goals

Progress Made

Blockers to Progress/Issues